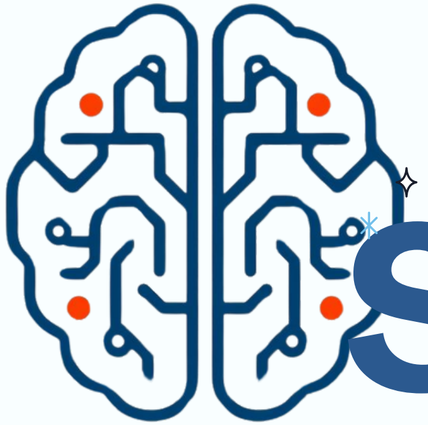
AI IN PRACTICE COURSE PROJECT



# SMART STUDY Assistant

Course
**AI in Practice**

Student
**Roya Abaste**

# Problem Statement

- Students face overwhelming amounts of study material including lecture notes, textbooks, and research papers that require significant time and effort to process effectively.

- Manual summarizing and note-taking is extremely time-consuming, leaving less time for actual learning and understanding of key concepts.

# Project Goal

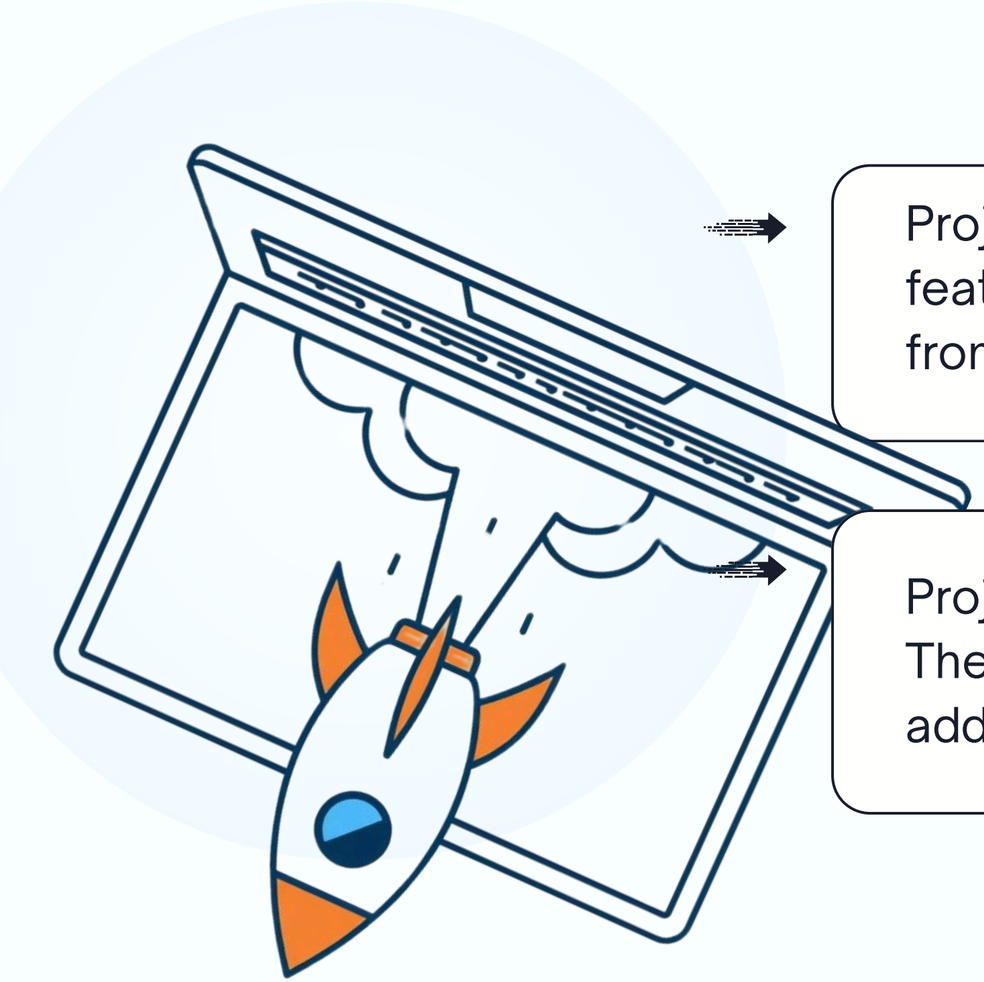
### Build AI Study Assistant

Create an intelligent system that converts PDF files into structured learning content. The assistant processes uploaded documents and transforms them into digestible study materials.

### Generate Learning Content

Automatically produce concise summaries, extract key points, and generate practice questions from study materials to enhance the learning experience.

# Current Stage



Project is in MVP stage (basic working version). Main features include PDF upload functionality, text extraction from documents, and AI summarization using Groq API.

Project is available on GitHub for collaboration and review. The foundation is set for future enhancements and feature additions.

# Project Overview

**Step 1**

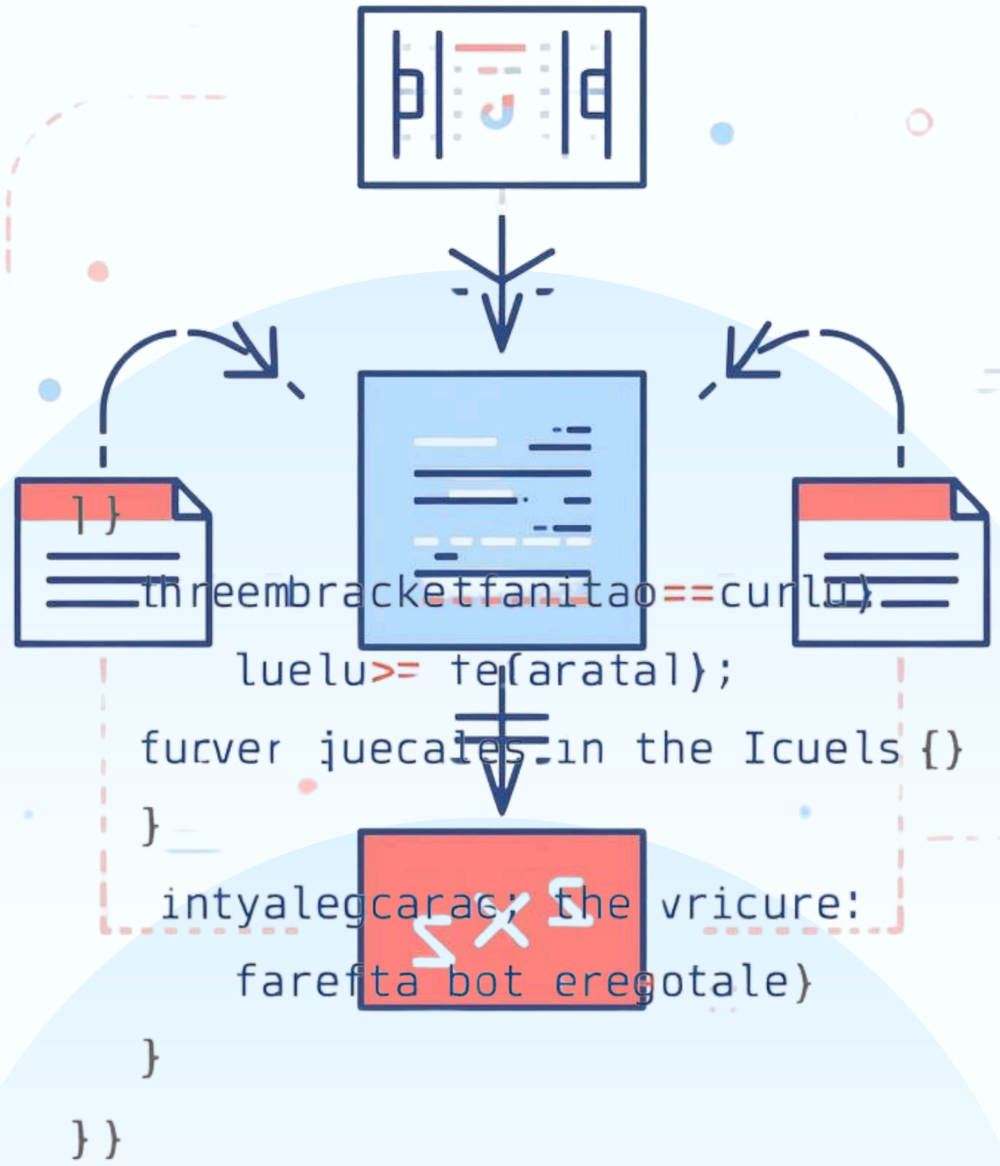
User uploads PDF file through the web interface. The system accepts standard PDF documents containing study materials.

**Step 2**

System extracts text from the PDF and sends it to the AI service via Groq API for intelligent processing.

**Step 3**

AI generates structured learning content: a concise summary, key points for review, and practice questions.



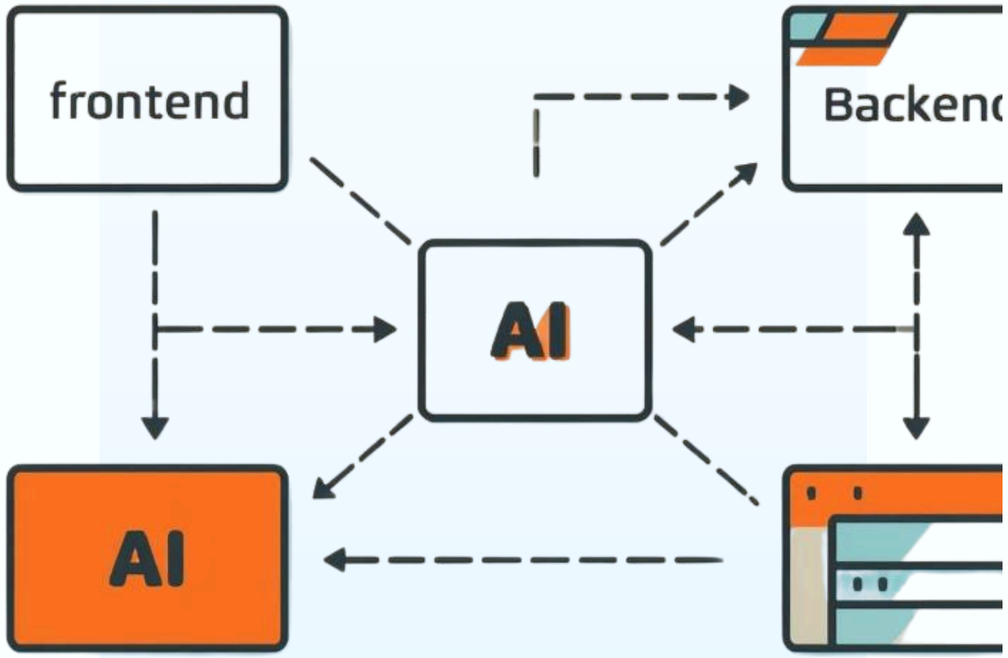

# AI Integration

Uses Groq API for intelligent text processing. The prompt creates structured JSON output:

- "summary": Condensed overview
- "keyPoints": Array of main concepts
- "questions": Array of study questions

AI transforms raw PDF text into organized, actionable learning content.

# System Design




- Frontend: Next.js
- Backend: API Routes
- PDF Processing: pdfjs
- AI Service: Groq API
- Database: Firebase
- Deployment: Vercel

- System Flow:
- Upload PDF
- Extract Text
- AI Processing
- Show Results

# Tools Used

Next.js
Groq API
pdfjs
Firebase
(future database + authentication)
Vercel (deployment)
GitHub

# Future Work & Conclusion

Future Work: Add login system, use Firebase database, generate flashcards, improve AI prompts, and enhance UI/UX design for better user experience.

Conclusion: This project demonstrates how AI can help students learn more effectively by transforming PDFs into simple, useful study materials including summaries, key points, and questions.